

PE-EC802A — Mixed Signal Design

Semester VIII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes

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1. Data Converter Performance Metrics

Data converters form the bridge between the analog physical world and the digital processing domain. Their performance is specified by static and dynamic parameters.

1.1 Static Metrics: DNL and INL

Let Δ (or $1 \text{ LSB} = V_{FS}/2^N$) represent the ideal step size of the converter.

Differential Non-Linearity (DNL)

Differential Non-Linearity (DNL) measures the deviation of an actual step width from the ideal value of 1 LSB. For an ADC, it is the difference between the actual step width W_k and the ideal 1 LSB:

$$\text{DNL}_k = \frac{W_k - \text{LSB}}{\text{LSB}}$$

If DNL becomes less than -1 LSB , the step width becomes zero or negative, leading to **missing codes** in ADCs. A system is guaranteed monotonic if $\text{DNL} > -1 \text{ LSB}$.

Integral Non-Linearity (INL)

Integral Non-Linearity (INL) measures the deviation of the actual transfer function characteristics from the straight line (ideal transfer curve) connecting the endpoints (zero-scale and full-scale). It represents the cumulative sum of DNL errors:

$$\text{INL}_k = \sum_{i=1}^k \text{DNL}_i$$

INL directly limits the harmonic distortion and linearity of the system.

1.2 Quantization Noise Derivation

Quantization is the process of mapping a continuous range of values to a discrete set. The difference between the continuous input x and its quantized representation $q(x)$ is the quantization error $e = x - q(x)$.

Assuming the input is active and spans many levels, the quantization error e is uniformly distributed between $-\Delta/2$ and $+\Delta/2$. The probability density function (PDF) is:

$$p(e) = \frac{1}{\Delta} \quad \text{for } -\frac{\Delta}{2} \leq e \leq \frac{\Delta}{2}$$

Step-by-Step Derivation of Quantization Noise Power (v_{qn}^2):

1. The noise power (variance of the error) is:

$$P_{qn} = \sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} e^2 p(e) de = \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de$$

2. Evaluating the integral:

$$P_{qn} = \frac{1}{\Delta} \left[\frac{e^3}{3} \right]_{-\Delta/2}^{\Delta/2} = \frac{1}{3\Delta} \left(\frac{\Delta^3}{8} - \left(-\frac{\Delta^3}{8} \right) \right) = \frac{1}{3\Delta} \left(\frac{2\Delta^3}{8} \right) = \frac{\Delta^2}{12}$$

Quantization Noise Power

$$P_{qn} = \frac{\Delta^2}{12}$$

3. For a sinusoidal input signal of peak amplitude $V_p = V_{FS}/2 = 2^N \Delta/2$, its signal power is:

$$P_{sig} = \frac{V_p^2}{2} = \frac{2^{2N} \Delta^2}{8}$$

4. The Signal-to-Quantization-Noise Ratio (SQNR) is:

$$\text{SQNR} = \frac{P_{sig}}{P_{qn}} = \frac{2^{2N} \Delta^2 / 8}{\Delta^2 / 12} = \frac{12}{8} 2^{2N} = 1.5 \times 2^{2N}$$

5. In decibels (dB):

Ideal SQNR Equation

$$\text{SQNR (dB)} = 10 \log_{10}(1.5) + 20N \log_{10}(2) = 1.76 + 6.02N$$

where N is the resolution of the converter in bits.

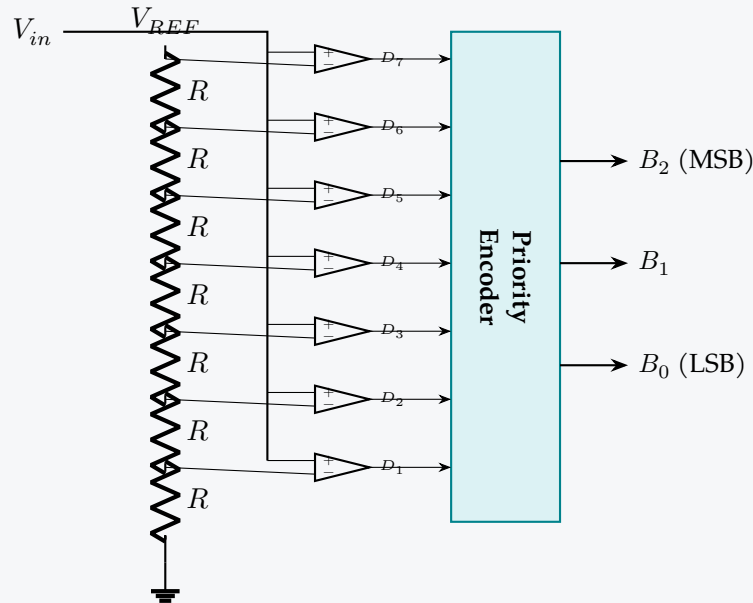
2. Analog-to-Digital Converter (ADC) Architectures

ADCs convert continuous voltage inputs to digital codes. The choice of architecture depends on the target speed and resolution.

2.1 1. Flash ADC (Ultra High Speed)

A Flash ADC uses $2^N - 1$ comparators biased by a resistor ladder divider to sample the input in parallel.

3-Bit Flash ADC Block Diagram



- **Working:** The resistor ladder sets $2^N - 1$ reference voltages. The comparators simultaneously compare V_{in} to these references, producing a thermometer code. The priority encoder translates this thermometer code to an N -bit binary code.
- **Pros:** Conversion finishes in a single clock cycle (highest speed, GHz range).
- **Cons:** Component count grows exponentially ($2^N - 1$ comparators). Power and input capacitance grow heavily, limiting resolution to 8 bits.

2.2 Successive Approximation Register (SAR) ADC (Moderate Speed/Resolution)

The SAR ADC utilizes a binary search algorithm to resolve the input voltage bit-by-bit over N clock cycles.

- **Working:** The Sample-and-Hold circuit holds V_{in} . The SAR logic sets the MSB of the DAC to 1. The comparator compares V_{in} to V_{DAC} . If $V_{in} > V_{DAC}$, the MSB remains 1; otherwise, it is reset to 0. The process repeats for the next bit until all N bits are resolved.
- **Pros:** Low power, minimal analog hardware (only 1 comparator, 1 DAC, and control logic).
- **Cons:** Requires N clock cycles per conversion, limiting speed.

2.3 Dual-Slope Integrator ADC (High Accuracy/Low Speed)

A dual-slope ADC integrates the input voltage for a fixed time T_1 , then de-integrates using a reference voltage V_{REF} until the integrator output returns to zero.

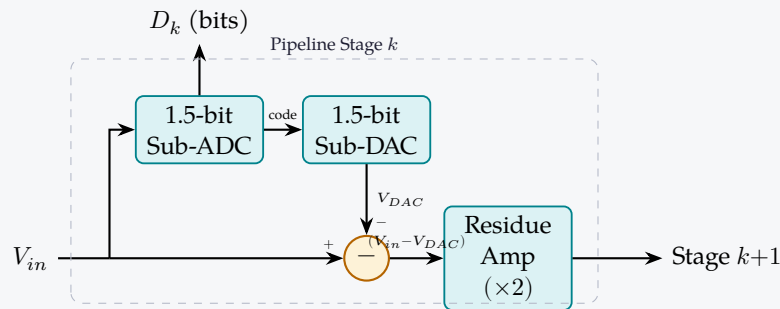
- **Working:** The integrator charges up with current $I_1 = V_{in}/R$ for 2^N clock cycles. Then, the input is switched to $-V_{REF}$, discharging the integrator back to 0. A counter measures the discharge time T_2 .
- **Equation:** The peak integrator voltage is $V_p = \frac{V_{in}}{RC} T_1 = \frac{V_{REF}}{RC} T_2 \Rightarrow V_{in} = V_{REF} \frac{T_2}{T_1}$.
- **Pros:** Outstanding noise rejection (integrates thermal noise out). The accuracy is independent of clock frequency and component values (R and C).

- **Cons:** Extremely slow (2^{N+1} clock cycles per sample). Used in digital multimeters.

2.4 4. Pipeline ADC (High Speed, Medium-High Resolution)

A Pipeline ADC achieves high throughput by processing multiple samples concurrently across a chain of stages. Each stage resolves a coarse set of bits, computes the remaining error (residue), amplifies it by $\times 2$, and passes it to the next stage.

Pipeline ADC: 1.5-Bit Stage Architecture



Working (1.5-bit/stage):

1. **Sub-ADC:** A 2-comparator flash resolves the MSBs using three levels (+1, 0, -1) compared against $\pm V_{REF}/4$.
 2. **Sub-DAC:** The coarse code is converted back to an analog level V_{DAC} .
 3. **Residue Amplifier (MDAC):** Computes $V_{res} = 2(V_{in} - V_{DAC})$, scaling the residue to fill the full input range of the next stage.
 4. **Pipelining:** While Stage 1 processes sample $n + 1$, Stage 2 concurrently processes the residue of sample n . This gives one output code per clock cycle after an initial latency of N_{stages} cycles.
 5. **Digital Error Correction:** The 0.5-bit redundancy allows the back-end logic to correct comparator offset errors up to $\pm V_{REF}/4$, relaxing hardware specifications significantly.
- **Pros:** High throughput (10 Msps–1 Gsps), excellent resolution (10–14 bits).
 - **Cons:** Multi-cycle pipeline latency; MDAC gain accuracy critically limits INL.

2.5 5. Hybrid ADC Structures

Hybrid ADCs merge two architectures to combine the speed strength of one with the resolution or power efficiency of the other.

Common Hybrid ADC Topologies

- **Flash-SAR Hybrid:** A coarse Flash ADC resolves the top M bits in a single cycle; a SAR ADC then refines the residue over $N - M$ cycles. Reduces comparator count from $2^N - 1$ (full Flash) to $2^M + N - M$, saving power while maintaining near-Flash speed.
- **Pipelined-SAR (SAR-assisted Pipeline):** Each pipeline stage employs a short SAR loop (4–6 bits) instead of the traditional 1.5-bit MDAC. This reduces the total number of stages and relaxes the MDAC gain-bandwidth requirement. Dominant in 10–14-bit,

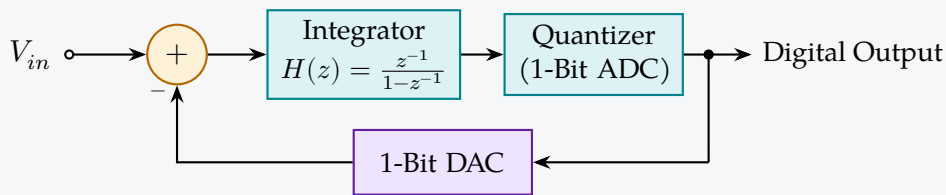
50–500 Msps ADCs in modern scaled CMOS nodes.

- **Incremental $\Delta\Sigma$ ADC:** Resets its accumulator after each conversion cycle, combining $\Delta\Sigma$ noise-shaping with discrete Nyquist-rate output. Used in precision multi-channel sensor readout (industrial, medical, weighing systems).

2.6 6. Delta-Sigma ($\Delta\Sigma$) ADC (Oversampling & Noise Shaping)

Delta-Sigma ADCs achieve ultra-high resolution (16–24 bits) by combining **Oversampling** and **Noise Shaping**.

First-Order Delta-Sigma Modulator Block Diagram



Oversampling: The input is sampled at $f_s \gg 2f_{max}$. The Oversampling Ratio is $OSR = f_s / (2f_{max})$. Oversampling spreads the same quantization noise power $\Delta^2/12$ over a wider bandwidth $f_s/2$, reducing the noise spectral density in the signal band.

Noise Shaping: By placing the quantizer inside a feedback loop with an integrator, the transfer function shapes the quantization noise. The output in the Z-domain is:

$$Y(z) = X(z)z^{-1} + E(z)(1 - z^{-1})$$

This shows that the input signal $X(z)$ is passed with a simple delay (Signal Transfer Function $STF(z) = z^{-1}$), while the quantization noise $E(z)$ is high-pass filtered (Noise Transfer Function $NTF(z) = 1 - z^{-1}$). The noise is pushed out of the low-frequency signal band into high frequencies, where it is easily removed by a digital decimation filter.

3. Digital-to-Analog Converter (DAC) Architectures

3.1 1. R-2R Ladder DAC (Compact Layout)

Unlike binary-weighted resistor DACs (which require huge component ratios like $2^N : 1$), the R-2R ladder uses only two resistor values (R and $2R$), making matching very easy.

The output voltage is:

$$V_{out} = V_{REF} \sum_{i=1}^N \frac{b_i}{2^i}$$

where b_1 is the MSB and b_N is the LSB.

3.2 2. Current-Steering DAC (High Speed)

Current-steering DACs route currents from matched MOS current sources to the output node based on the digital input code.

- **Working:** An array of current sources (either binary-weighted or thermometer-coded) is connected to differential switches. Thermometer coding is preferred for high resolution

because it guarantees monotonicity and minimizes output glitches, though it requires a thermometer decoder.

- **Pros:** Extremely fast (up to GHz range), can drive resistive loads (like $50\ \Omega$ coaxial cables) directly.

Quick Revision — Module 3

Data Converters | PE-EC802A

Key Formulas at a Glance

Quantization Error Power	$P_{qn} = \frac{\Delta^2}{12}$
Ideal SQNR (dB)	$SQNR = 6.02N + 1.76 \text{ dB}$
Oversampling Ratio (OSR)	$OSR = \frac{f_s}{2f_{max}}$
1st-Order Modulator NTF	$NTF(z) = 1 - z^{-1}$
2nd-Order Modulator NTF	$NTF(z) = (1 - z^{-1})^2$
Flash Comparator Count	$N_{comp} = 2^N - 1$
Dual-Slope conversion time	$T_{conv} \approx 2^{N+1}T_{clk}$

One-Line Definitions

- **DNL:** The deviation of an actual step size from the ideal 1 LSB step width.
- **INL:** The cumulative deviation of the actual transfer curve from the ideal straight line.
- **Oversampling:** Sampling a signal at a rate significantly higher than the Nyquist rate.
- **Noise Shaping:** Pushing quantization noise from the signal band to higher frequencies using feedback loops.
- **Monotonicity:** A converter characteristic where the output always increases or stays constant as input increases.
- **Thermometer Code:** A unary code format representing value K by K consecutive ones (e.g., $3 \rightarrow 0000111$).
- **Missing Codes:** An ADC defect where certain output codes are skipped, caused by $DNL < -1 \text{ LSB}$.
- **Pipeline ADC:** A high-speed converter using chained 1.5-bit stages with residue amplification, enabling one output per clock cycle.
- **Hybrid ADC:** A converter combining two architectures (e.g., Flash-SAR, Pipelined-SAR) to trade off speed, resolution, and power.

ADC Architecture Comparison at a Glance

ADC Type	Speed	Resolution	Power	Application
Flash	GHz range	≤ 8 bit	Very High	Oscilloscopes, SerDes
SAR	< 100 Msps	8–18 bit	Very Low	Sensors, IoT, MCUs
Dual-Slope	< 100 sps	20+ bit	Low	Multimeters
Pipeline	10–1000 Msps	10–14 bit	Medium	WLAN, Video, RADAR
$\Delta\Sigma$	< 10 Msps	16–24 bit	Medium	Audio, Precision
Hybrid	Varies	Varies	Low–Medium	Imaging, Comm. IC

Frequently Asked / Viva Questions

Q1. Derive the quantization noise power of an ideal ADC.

For a quantization step size Δ , the error e is uniformly distributed between $-\Delta/2$ and $\Delta/2$ with PDF $1/\Delta$. The noise power is the variance: $P_{qn} = \int_{-\Delta/2}^{\Delta/2} e^2 (1/\Delta) de = [e^3/3\Delta]_{-\Delta/2}^{\Delta/2} = \Delta^2/12$.

Q2. What causes missing codes in an ADC, and how is it defined mathematically?

Missing codes occur when the transition step width of a digital code is zero or negative. This happens when the Differential Non-Linearity (DNL) at that code is less than or equal to -1 LSB.

Q3. Explain the trade-offs of Flash ADC architecture.

Flash ADC is the fastest architecture because it performs parallel comparisons in one clock cycle. However, it requires $2^N - 1$ comparators. This causes exponential growth in area, power dissipation, and input capacitance, limiting the resolution to 8 bits.

Q4. How does a Delta-Sigma ADC achieve very high resolution?

By combining oversampling (which spreads quantization noise over a wider band) and noise shaping (which uses a loop filter to push noise to high frequencies). A digital low-pass decimation filter then filters out the out-of-band noise, leaving a high-resolution signal.

Q5. Why is a dual-slope ADC highly accurate, and where is it used?

Its accuracy depends only on the ratio of charging and discharging times (T_2/T_1), which is independent of the absolute values of R , C , and clock frequency. It is used in digital multimeters where accuracy is critical and speed is not.

Q6. Compare SAR and Flash ADCs in terms of speed, area, and power.

Flash ADC converts in 1 clock cycle but requires $2^N - 1$ comparators (huge area and power). SAR ADC takes N clock cycles but requires only 1 comparator and 1 DAC, resulting in very low power and compact area.

Frequently Asked / Viva Questions (contd.)**Q7. What is the benefit of the R-2R ladder DAC over a binary-weighted resistor DAC?**

A binary-weighted resistor DAC requires a very wide range of resistor values ($2^N : 1$), which is difficult to match on-chip. The R-2R ladder DAC uses only two resistor values (R and $2R$), ensuring excellent matching and layout compacting.

Q8. Explain the function of the priority encoder in a Flash ADC.

The comparators output a thermometer code (e.g. 0000111). The priority encoder identifies the highest active input line and converts that thermometer code into standard binary output (e.g. 011).

Q9. What is the Noise Transfer Function (NTF) of a first-order Delta-Sigma modulator?

The NTF is a high-pass filter: $\text{NTF}(z) = 1 - z^{-1}$. In the frequency domain, it is $2 \sin(\omega T_s/2)$, which attenuates noise near DC and amplifies it at high frequencies.

Q10. What is the difference between DNL and INL?

DNL is the local step size error from the ideal 1 LSB step width. INL is the global deviation of the actual transfer function from the ideal straight line, equal to the cumulative sum of the DNL errors.